

Suresh Nayak

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Staff Software Engineer

Expertise: Expertise in Distributed Systems, AI/ML Infrastructure, and Scalable Platform Engineering

Professional Experience

Arrcus - Platform/Forwarding Team (San Jose, CA)

Sr Software Engineer

May 2023 - Present

- Spearheaded a vendor-agnostic firmware upgrade system, reducing operational complexity and enabling rapid iteration across hardware platforms.
- Integrated Linux IMA into ArcOS, enabling signed-binary enforcement and strengthening system **security**.
- Designed system architecture improvements that minimized attack surface via **least-privilege capability enforcement**, directly strengthening production security and stability.
- **Architected and owned migration of ArcOS process management to systemd**, delivering higher reliability, fault recovery, and scalable orchestration for production systems, **directly contributing to the stability of AI/ML networking platforms**
- **Led cross-functional debugging and scaling efforts** (memory leaks, performance bottlenecks), ensuring resilience in customer-facing AI/ML networking platforms.
- **Arcos for AI/ML infrastructure development:**
 - Designed and coded the platform layer of ArcOS to bridge the ArcOS application with the Spectrum 3 SDK.
 - Developed and implemented features such as BOND/LAG, L2, and L3 interfaces, port mirroring, and bridging.
- Mentored and coached an engineer on the platform layer, enhancing team capabilities and knowledge transfer
- **Broadcom Tomahawk 5 Platform Bring-up for AI/ML Networking:**
 - Successfully brought up components to provide Access control, Quality of service, and CoPP on the Tomahawk 5 platform.
 - Designed and developed Counter infrastructure for the XGS platform, enabling all processes to leverage this feature without understanding platform-specific intricacies
 - Debugged memory corruption and leak issues in the platform layer, improving system stability and performance

Cisco Systems Software Engineer - Software Engineer

Aug 2018 - Aug 2021

- Redesigned a critical software mechanism for monitoring process liveness in Cisco XR routers, enabling automatic recovery of hung processes and **enhancing the reliability of distributed network systems**.
- Optimized high-availability components during product release cycles, reducing workload overhead by 80% through a meticulously devised feature, enhancing efficiency and reliability.
- Led the migration of software components from GCC to Clang Compiler, enforcing inter-component tight coupling and achieving a reduction in bug count by nearly 10%.

University of Colorado Boulder

Graduate Teaching Assistant

Aug 2021– May 2023

- Led recitation classes, graded interviews, and conducted office hours for the "Computer Systems" course, effectively supporting and mentoring a class of 379 students

Education

University of Colorado Boulder (Boulder, CO)

M.S. in Computer Science, **GPA: 3.76**

2023

RV University (Bangalore, India)

B.E. in Information Science, **GPA: 8.6**

2018

Projects

ShelfHelp – Vision-Independent Robotic Manipulation: Delivered deep learning–based item recognition system (YOLOv5, 93% accuracy) enabling accessibility-driven robotic manipulation; published at AAMAS.

Autonomous Goal Recognition System: Built an adaptive navigation framework (POMDP-based) capable of real-time human intent inference, demonstrating scalable decision-making for complex environments.

AAMAS:[ShelfHelp: Empowering Vision-Independent Manipulation with a Robotic Cane](#)

Technical skills: Python, C/C++, Go, SQL, HTML/CSS, Assembly, MySQL, NumPy, Pandas, TensorFlow, Pytorch, TF Light, Media pipe, Git, Docker, GitHub Actions, Postman, Kubernetes. Robotic Operating System (ROS), Google Cloud Platforms, CI/CD